Unit 3 Handout: Probability

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Example 1

Suppose we have a 6-sided die and roll the die one time.

$E = \{\text{an even number appears}\}$
$F = \{\text{a number less than or equal to 4 appears}\}$

Find $P(E)$, $P(F)$, $P(E \cap F)$, $(P(E \cup F)$, and $P(\overline{E})$.

Example 2

Suppose, we select a freshman at random at CWRU.

$P(\text{enrolled in Calculus}) = P(C) = .3$
$P(\text{enrolled in History}) = P(H) = .5$
$P(\text{enrolled in both classes}) = P(H \cap C) = .2$

Draw a Venn diagram & find the following:

(a) $P(H \cup C)$

(b) $P\left(\overline{H \cup C}\right)$

(c) $P(H \cap \overline{C})$

(d) $P(\overline{H} \cup C)$

(e) $P(C \mid H)$

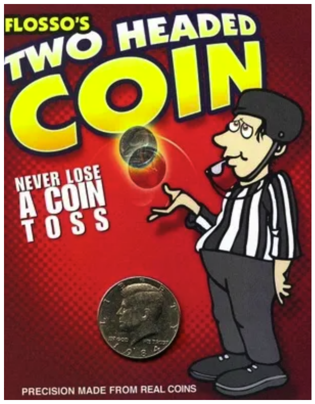
(f) Are C and H independent? Justify your answer.

(g) Are C and H mutually exclusive?

(h) H and ________________ are mutually exclusive.

Example 3

Suppose John is flipping a 2-sided coin. If he gets a tail on the first flip, he stops the process. However, if he gets a head on the first flip, he flips the coin once more.



(a) Draw a tree diagram.

(b) Find $P(\text{Head appears} \mid \text{Tail appears})$.

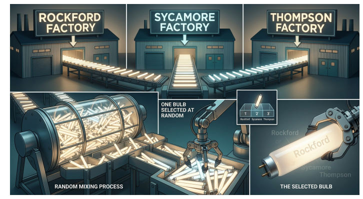
Example 4

Suppose 1% of the people on an island have Disease X. A test for the disease is 99% accurate.

(a) Given that someone on the island tests positive, what is the probability that they actually have the disease?

(b) Given that someone tests positive three times in a row, what is the probability that they actually have the disease?

Example 5



A company makes fluorescent light bulbs in three factories: Rockford, Sycamore, and Thompson. One bulb is selected at random.

Suppose, 70% are made in Rockford: $P(R) = .70$
20% are made in Sycamore: $P(S) = .20$
10% are made in Thompson: $P(T) = .10$

Let, $W = \{\text{a bulb is still working after 10,000 hours of use}\}$.

Find $P(W \mid S)$, $P(W \mid T)$, $P(W)$, $P(S \cup \overline{W})$, $P(W \mid T), and$ $P(T \mid W)$.

Example 6

Suppose a basket contains 10 balls: 7 red, 2 green, and 1 black.

Three balls will be drawn one at a time at random.

Considering both **with replacement** and **without replacement**, find the following probabilities:

$P(RRR)$, $P(BGB)$, $P(\text{no black draws}), \textit{and}$ $P(\text{at least one green})$.

Example 7: The Famous Birthday Problem

Would you be surprised if two people in this room shared the same birthday?
Find $P(\text{at least one shared birthday})$.

Example 8

Suppose we are rolling two fair dice. One die is red and the other is blue. There are 36 equally likely outcomes in the sample space.

(red $= 1$, blue $= 1$)
(red $= 2$, blue $= 1$)
(red $= 3$, blue $= 1$)
$\vdots$
(red $= 5$, blue $= 6$)
(red $= 6$, blue $= 6$)

The table below has 36 cells corresponding to these 36 outcomes. The sum corresponding to each outcome appears in the appropriate cell.

	1	**2**	**3**	**4**	**5**	**6**
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Find each of the following probabilities:

(a) $P(\text{sum} = 7)$

(b) $P(\text{sum} = 2)$

(c) $P(\text{blue} = 6 \cap \text{sum} \geq 11)$

(d) $P(\text{sum is even} \cup \text{red} = 1)$

(e) $P(\text{red} = 4 \mid \text{sum} = 10)$

(f) $P(\text{blue} = 1 \mid \text{blue} < \text{red})$

	1	**2**	**3**	**4**	**5**	**6**
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

(g) $P(\text{sum is odd} \mid \text{red} < 4)$

Identify whether the following events are **independent**, **mutually exclusive**, or **neither**. Justify your answer.

Red $= 5$ and sum $= 9$

Checking:

Are they mutually exclusive?

$$P(\text{red} = 5 \cap \text{sum} = 9) =$$

Are they independent?

$$P(\text{red} = 5 \cap \text{sum} = 9) \stackrel{?}{=} P(\text{red} = 5) \times P(\text{sum} = 9)$$

$P(\text{red} = 5) =$

$P(\text{sum} = 9) =$

$P(\text{red} = 5) \times P(\text{sum} = 9) =$